

Assignment 1

1. What is the difference between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

- Native: Build only one specific operating system using dedicated tools or language. Like iOS uses Swift, while Android uses Kotlin and Java
- Hybrid: Built using web technologies like HTML5, CSS, and JavaScript, then placed inside a native app shell or webview. The drawback of using hybrid is that it is slower than native
- Cross-platform: Uses a single shared codebase written in frameworks like React Native or Flutter to target both iOS and Android.
- The most commonly used is Cross-platform because it can support Android and IOS by using Flutter and React native because they are fast to develop and easy to maintain

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

- The Flutter Ecosystem is a collection of libraries, frameworks, and tools that work together to help developers build, test, and deploy apps using Flutter, more like buy-one-get-one for developers

3. What is the Software Development Life Cycle(SDLC), and why is it important in building successful applications?

- SDLC is a systematic approach to developing software, from initial planning through ongoing maintenance. By following its structured phases, development teams can build reliable, secure, and high-quality applications while reducing risks
- This is important cause it provides a clear development process to ensure the following phase is completed before proceeding to the next step, also

improves software quality through testing and quality assurance, and enhances project management through better planning, scheduling, and resource allocation

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

- Flutter development practices and the Software Development Life Cycle (SDLC) work together by combining a structured development process with Flutter's tools and features. While the SDLC provides the roadmap for building an application, Flutter provides the technology needed to efficiently develop, test, and deploy it.

5. Why should developers understand these concepts before starting an application project?

- Developers should understand concepts such as Native, Hybrid, and Cross-platform development, the Flutter ecosystem, and the Software Development Life Cycle (SDLC) before starting an application project because these concepts help them make informed decisions, work efficiently, and build high-quality software.